

JAVA PATTERN

— P R O G R A M S —

BASIC TO ADVANCED PATTERNS

1. RIGHT TRIANGLE

```

*
* *
* * *
* * * *
* * * * *
    
```

CODE

```

int n = 5;
for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= i; j++) {
        System.out.print("* ");
    }
    System.out.println();
}
    
```

2. NUMBER TRIANGLE

```

1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
    
```

CODE

```

int n = 5;
for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= i; j++) {
        System.out.print(j + " ");
    }
    System.out.println();
}
    
```

3. REVERSE TRIANGLE

```
* * * * *
* * * *
* * *
* *
*
```

CODE

```
int n = 5;
for (int i = n; i >= 1; i--) {
    for (int j = 1; j <= i; j++) {
        System.out.print("* ");
    }
    System.out.println();
}
```

4. REVERSE NUMBER TRIANGLE

```
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1
```

CODE

```
int n = 5;
for (int i = n; i >= 1; i--) {
    for (int j = 1; j <= i; j++) {
        System.out.print(j + " ");
    }
    System.out.println();
}
```



5. PYRAMID PATTERN

```

      *
    * * *
  * * * * *
* * * * * * *
* * * * * * * *
  
```

CODE

```

int n = 5;
for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= n - i; j++) {
        System.out.print(" ");
    }
    for (int j = 1; j <= 2 * i - 1; j++) {
        System.out.print("*");
    }
    System.out.println();
}
  
```

6. INVERTED PYRAMID PATTERN

```

* * * * * * * *
 * * * * * * *
  * * * * *
   * * *
    *
  
```

CODE

```

int n = 5;
for (int i = n; i >= 1; i--) {
    for (int j = 1; j <= n - i; j++) {
        System.out.print(" ");
    }
    for (int j = 1; j <= 2 * i - 1; j++) {
        System.out.print("*");
    }
    System.out.println();
}
  
```



7. DIAMOND PATTERN

```

      *
    * * *
  * * * * *
* * * * * *
  * * * *
    * * *
      *
  
```

CODE

```

int n = 5;
for(int i = 1; i <= n; i++) {
    for(int j = 1; j <= n - i; j++) {
        System.out.print(" ");
    }
    for(int j = 1; j <= 2 * i - 1; j++) {
        System.out.print("* ");
    }
    System.out.println();
}
for(int i = n - 1; i >= 1; i--) {
    for(int j = 1; j <= n - i; j++) {
        System.out.print(" ");
    }
    for(int j = 1; j <= 2 * i - 1; j++) {
        System.out.print("* ");
    }
    System.out.println();
}
  
```

8. HOLLOW PYRAMID PATTERN

```

      *
    *   *
  *     *
*       *
  *     *
    *   *
      *
  
```

CODE

```

int n = 5;
for(int i = 1; i <= n; i++) {
    for(int j = 1; j <= n - i; j++) {
        System.out.print(" ");
    }
    for(int j = 1; j <= 2 * i - 1; j++) {
        if(j == 1 || j == 2 * i - 1 || i == n)
            System.out.print("* ");
        else
            System.out.print(" ");
    }
    System.out.println();
}
  
```


9. BUTTERFLY PATTERN

```

* * * * *
* * * * *
* *       * *
*         *
*         *
* *       * *
* * * * *
* * * * *
    
```

CODE

```

int n = 4;
for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= i; j++)
        System.out.print("* ");
    for (int j = 1; j <= 2 * (n - i); j++)
        System.out.print(" ");
    for (int j = 1; j <= i; j++)
        System.out.print("* ");
    System.out.println();
}
for (int i = n; i >= 1; i--) {
    for (int j = 1; j <= i; j++)
        System.out.print("* ");
    for (int j = 1; j <= 2 * (n - i); j++)
        System.out.print(" ");
    for (int j = 1; j <= i; j++)
        System.out.print("* ");
    System.out.println();
}
    
```

10. X PATTERN

```

*           *
 *         *
  *       *
   *     *
    *   *
     * *
      *
     * *
    *   *
   *     *
  *       *
 *         *
*           *
    
```

CODE

```

int n = 4;
for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= n; j++) {
        if (i == j || i + j == n + 1)
            System.out.print("* ");
        else
            System.out.print(" ");
    }
    System.out.println();
}
    
```

11. HOURGLASS PATTERN

```

* * * * *
  * * *
    *
  * * *
* * * * *
  
```

CODE

```

int n = 5;
// Upper part
for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= 2 * n - 1; j++) {
        if (j >= n - i + 1 && j <= n + i - 1)
            System.out.print("* ");
        else
            System.out.print(" ");
    }
    System.out.println();
}
// Lower part
for (int i = n - 1; i >= 1; i--) {
    for (int j = 1; j <= 2 * n - 1; j++) {
        if (j >= n - i + 1 && j <= n + i - 1)
            System.out.print("* ");
        else
            System.out.print(" ");
    }
    System.out.println();
}
  
```

12. HOLLOW DIAMOND PATTERN

```

      *
     * *
    *   *
   *     *
  * * * *
 *       *
*         *
 *       *
  *     *
   *   *
    * *
     *
      *
  
```

CODE

```

int n = 5;
// Upper part
for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= n; j++) {
        if (j == 1 || j == i || i == n)
            System.out.print("* ");
        else
            System.out.print(" ");
    }
    System.out.println();
}
// Lower part
for (int i = n - 1; i >= 1; i--) {
    for (int j = 1; j <= n; j++) {
        if (j == 1 || j == i || i == 1)
            System.out.print("* ");
        else
            System.out.print(" ");
    }
    System.out.println();
}
  
```

MOSTLY ASKED PATTERNS IN JAVA

13. RIGHT TRIANGLE

```

*
* *
* * *
* * * *
* * * * *
```

CODE

```

int n = 5;
for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= i; j++) {
        System.out.print("* ");
    }
    System.out.println();
}
```

14. INVERTED RIGHT TRIANGLE

```

* * * * *
* * * *
* * *
* *
*
```

CODE

```

int n = 5;
for (int i = n; i >= 1; i--) {
    for (int j = 1; j <= i; j++) {
        System.out.print("* ");
    }
    System.out.println();
}
```


15. SQUARE PATTERN

```
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
```

CODE

```
int n = 5;
for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= n; j++) {
        System.out.print("* ");
    }
    System.out.println();
}
```

16. RHOMBUS PATTERN

```
      *
     * * *
    * * * * *
     * * *
      *
```

CODE

```
int n = 3;
for (int i = 1; i <= 2 * n - 1; i++) {
    int spaces = Math.abs(n - i);
    for (int j = 1; j <= spaces; j++) {
        System.out.print(" ");
    }
    for (int j = 1; j <= 2 * (n - spaces) - 1; j++) {
        System.out.print("* ");
    }
    System.out.println();
}
```

17. ALPHABET TRIANGLE

```

      A
    A  B
  A  B  C
A  B  C  D
A  B  C  D  E
  
```

CODE

```

int n = 5;
char ch;
for (int i = 1; i <= n; i++) {
    ch = 'A';
    for (int j = 1; j <= i; j++) {
        System.out.print(ch + " ");
        ch++;
    }
    System.out.println();
}
  
```

18. INVERTED ALPHABET TRIANGLE

```

A  B  C  D  E
  A  B  C  D
    A  B  C
      A  B
        A
  
```

CODE

```

int n = 5;
char ch;
for (int i = n; i >= 1; i--) {
    ch = 'A';
    for (int j = 1; j <= i; j++) {
        System.out.print(ch + " ");
        ch++;
    }
    System.out.println();
}
  
```

MOSTLY ASKED NUMBER PATTERNS IN JAVA

19. NUMBER TRIANGLE

```

      1
     1 2
    1 2 3
   1 2 3 4
  1 2 3 4 5
  
```

CODE

```

int n = 5;
for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= i; j++) {
        System.out.print(j + " ");
    }
    System.out.println();
}
  
```

20. PALINDROME NUMBER PATTERN

```

      1
     1 2 1
    1 2 3 2 1
   1 2 3 4 3 2 1
  1 2 3 4 5 4 3 2 1
  
```

CODE

```

int n = 5;
for (int i = 1; i <= n; i++) {
    // first part
    for (int j = 1; j <= i; j++) {
        System.out.print(j + " ");
    }
    // second part
    for (int j = i - 1; j >= 1; j--) {
        System.out.print(j + " ");
    }
    System.out.println();
}
  
```



21. SQUARE PATTERN

```
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
```

CODE

```
int n = 5;
for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= n; j++) {
        System.out.print("* ");
    }
    System.out.println();
}
```

22. RECTANGLE PATTERN

```
* * * * *
* * * * *
* * * * *
* * * * *
```

CODE

```
int rows = 4, cols = 6;
for (int i = 1; i <= rows; i++) {
    for (int j = 1; j <= cols; j++) {
        System.out.print("* ");
    }
    System.out.println();
}
```

